Lars Crawford, Ph.D.

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Neural engineer turned software engineer, now leading AI adoption at Blockchain.com. PhD in Bioengineering (UW, Ratner Lab): in vitro and in vivo neural-interface research, two peer-reviewed papers, two patent filings; earlier, computational neuroscience and electrophysiology. Four years shipping production software, most recently on an AI platform used by tens of thousands daily. Since August, I've designed the context, test gates and guardrails for Claude Code agents that wrote nearly all of a shipped app. I want to put Claude to work in life-science research.

EXPERIENCE

Blockchain.com — Technical Product Lead, AI Transformation · London (hybrid) · May 2026 – present

- Lead AI adoption across the company: set strategy, onboard and manage AI vendors for coding and workflow automation, and design and build internal AI tools and bespoke plugins; near-term focus is engineering throughput.

Blockchain.com — Software Engineer (Web3 team), June AI · London · Sep 2025 – May 2026

- Joined June AI (askjune.ai) when it was a chat app with a model picker; full-stack engineer as it grew into a privacy-first, multi-model AI platform used by tens of thousands daily.
- Built side-by-side multi-model chat by rebuilding SSE stream handling in the client (the AI SDK's useChat couldn't run concurrent streams on one page); one model then synthesizes the answers into a cross-checked summary.
- Core contributor to end-to-end-encrypted chat sharing, still live and heavily used: chats are stored only on the user's device, so the browser encrypts, the server stores only ciphertext, and the key travels only in the link.
- Built client-side IndexedDB migrations that run on users' devices, against data the team could never see; when rolling deploys sent migrated clients to old instances, proposed blue-green releases with canaries, which SRE built.

Tribally — Software Engineer (contract), then Senior Software Engineer · London · Sep 2023 – Sep 2025

- Helped launch and grow tribally.games to 175k+ gamers and dozens of partner games.
- Led 2 frontend and 4 backend engineers on the Tribally DevKit Partner Portal, where partner studios set up their own integrations; built it with partner feedback and wrote content for its developer docs.
- Onboarded partner games and built gameplay-data ingestion APIs for 5+ of them across dozens of microservices.
- Built services that securely automate blockchain transactions on six networks: 80k+ transactions, 30k+ prizes sent.

Defenders of Lunacian Land — Lead Full-Stack Developer (part-time) · Jun 2022 – May 2025

- Built server-side auth, gameplay-data ingestion, role-based analytics for skill balancing, and payment processing that handled 10k+ payments and $100k+ in revenue, for one of the first projects in Axie Infinity's Builder's Program.

MultiModal Health — Cofounder, Director of Operations & Research · Seattle · 2015 – 2016

- Co-developed vHAB, a gamified stroke-rehabilitation device that integrates bio-signals from multiple sensors; designed and ran IRB-approved studies of patient motivation and engagement, recruiting patients and working directly with their therapists and physicians.

SELECTED AI PROJECTS

Immortal Kinematics — AI motion-capture dance app, built by an agent team I direct · Aug 2026 – present

- Designed the agent architecture that builds it: a scheduled Claude Code lead wakes every 3 hours with no memory, boots from a fixed set of repo files, delegates bounded tasks to up to three Opus subagents in isolated git worktrees, reviews every diff, and ships only what passes an automated test gate. Claude co-authored 2,416 of 2,593 commits in 8 weeks; I supply direction and taste, mostly by voice.
- Treated context as infrastructure: 270 architecture decision records and 241 self-contained task specs (goal, owned files, acceptance criteria, how to verify) are the shared memory for stateless agents across three Claude surfaces.
- Made the loop survive failures: subagents push after every step (a rule added after a lead session died mid-run) and never merge; a lead blocked on me switches to the most neglected workstream.
- Built the one-command test gate every agent change must pass (~137 steps: compile, tests, auth smoke test, build, headless desktop and phone screenshots). The lead's prompt forbids credential entry, holds spending and public posts for my yes, queues human-only questions and treats file and tool content as untrusted data.
- Shipped the app (FastAPI, Postgres, React): MediaPipe 33-joint pose extraction with multi-dancer tracking turns dance video into ranked clips and skeleton-scored challenges. Tests guard user data (one merge path, one deletion path); training consent is asked once, with an off switch and a right to object that purges exported copies.
- Pre-registered the challenge scorer's thresholds before it existed: 5/5 judged cases passed, with 20/20 mirror detection and body-size fairness. Judged a learned clip-picker's win over hand weights too small to claim (n = 43).
- Designed a human-in-the-loop clip improver: it mines my clip ratings, A/B-tests one bounded setting automatically and proposes a change only on a strict win; I accept or reject it.

- Ran a second agent loop on 3D-pose research: self-calibrated stereo capture supervises 2D-to-3D pose models (TCN, transformer) trained on a local GPU; 32 milestones, ties and losses included; model kept out of production.
- Wrote about it: "I built a dance app in a month by talking to it".

googolem — locally hosted, privacy-first agent (personal) · Feb – Aug 2026
- Built a LangGraph ReAct agent with ~15 tools (each wrapped in failure memory), background sub-agents, Mem0 long-term memory, RAG over my chat history (ChromaDB) and structured extraction into pydantic models.

mindmapper — a citation-gated research agent; literature curation in miniature · Aug – Sep 2026
- Wrote a Claude research-agent skill that curates a cited map of 173 AI models: a primary-source citation for every claim, no fabrication, a CI validator on every edit, and a pass counts as done only when validation and the build pass. An autonomous loop verified 166 of them and left the other 7 unverified rather than guess.

RESEARCH

University of Washington — PhD research, Ratner Lab · 2016 – 2021
- Thesis: *Toward a Wholly Organic, Immunomodulatory Neuroelectronic Interface.* Designed a new one-pot photochemical synthesis for soft, conductive hydrogels (PHEMA with PEDOT, graphene and carbon nanotubes) for long-term neural interfacing; co-inventor on the resulting patent application.
- Fabricated stand-alone hydrogel electrodes and characterized them mechanically, electrochemically and morphologically; tuned stiffness, porosity, biomolecule immobilization and degradability.
- Ran in vitro and in vivo studies: a mechanistic study of pore-size-dependent immunomodulation of macrophages and microglia, and immunohistochemical analysis of gliosis around implanted porous hydrogels.
- Designed precision-porous hydrogel scaffolds computationally and 3D-printed them by stereolithography; led interdisciplinary teams; published and presented internationally.

University of New South Wales, Poole-Warren Lab — Research intern · Sydney · Jan – Jul 2019
- Extended the one-pot synthesis to a PVA macromer system, showing the method is chemistry-agnostic.

UW Computational Neuroscience Program, Bair Lab · 2013 – 2014
- Built computational models of large-scale, biologically plausible functional maps in macaque primary visual cortex.

Center for Integrative Brain Research, Welsh Lab · Seattle · 2012 – 2013
- Ran whole-cell and dual whole-cell recordings of inferior-olive circuitry in ex vivo brain slices.

EDUCATION
- **Ph.D., Bioengineering and Nanotechnology & Molecular Engineering**, University of Washington, 2021.
- **B.S., Neurobiology**, University of Washington, 2014, cum laude; College and Departmental Honors.

PUBLICATIONS & PATENTS
- Crawford L, Wyatt M, Bryers J, Ratner B. "Biocompatibility Evolves: Phenomenology to Toxicology to Regeneration." *Advanced Healthcare Materials* 10(11):2002153 (2021).
- Zhen L, Darrow R, Chen N, Anant M, Gorantla L, Crawford L, Dryg I, Ratner B. "Soft, precision engineered porous, hydrogel scaffolds mechanically tailored toward applications in the central nervous system." *J Bioact Compat Polym* 39(6), 2024.
- Ratner BD, Crawford L. "One-pot photochemical synthesis methodology for conductive hydrogel fabrication." US Patent Application 17/357,887 (pub. US 2021/0403613 A1). Also: US Provisional 63/216,249, "Heterogeneous Unit Cells and Composites Containing the Same."
- Honors: 1st Place, Tissue Engineering SIG Award, Society for Biomaterials (2019); Best Poster, Neuroelectronic Interfaces Gordon Research Conference (2018); 1st Place, Center for Neurotechnology Tech Studio Competition (2014); Computational Neuroscience Training Grant (2013–14).

SKILLS
- **AI engineering:** Claude Code (scheduled agents, Opus subagents in worktrees, skills, test gates), Cowork; agent architectures, context engineering, prompt guardrails, research loops with citation rules, pre-registered benchmarks; LangGraph (tools, sub-agents, Mem0), RAG (ChromaDB), pydantic structured outputs, Vercel AI SDK, OpenRouter.
- **Software:** TypeScript, Python, React/Next.js; PostgreSQL, MongoDB, Redis; Kafka; Docker, Kubernetes, GCP.
- **Science:** hydrogel synthesis and characterization, cell culture, ELISA, immunohistochemistry, electrophysiology, 3D printing (SLA), in vitro/in vivo study design, computational neuroscience, IRB human-subjects research.